

ARM-450 — Glossary

Every term used across the ARM-450 documents, in plain language, with the number it actually took in this project.

1. The verification words

Gate

A single program that runs every check and gives one verdict. `preflight.py`. The idea is that you never have to remember which of forty checks you last ran — you run the gate and it tells you GO or NO-GO. Borrowed from manufacturing, where a "gate" is a point work cannot pass until it is signed off.

Check

One question with a yes/no answer and a stated limit. "Is the worst safety factor at least 3?" — not "is it strong?" A check you cannot fail is not a check.

Stage

A group of related checks. ARM-450 has eight, run in order — design geometry, FEA, creep and fatigue, stiffness, URDF, path tracing, simulation, end effectors. They are ordered so the cheap checks fail first: there is no point running a 4-minute collision sweep if a bolt hole is missing.

Gate passed / GO

Every hard check passed. In ARM-450 it means: the features physically exist in the CAD, nothing yields, the arm is stiff enough, the model matches the machine, the trajectory does not collide, and the ROS side will work on hardware.

What it does not mean: that the arm works. Nothing has been printed. GO means the design is self-consistent and inside its own limits — no more.

NO-GO

At least one hard check failed. The gate names it and stops. ARM-450 currently reads NO-GO on one item: assembled mass 1584 g against a 1450 g budget.

Hard check vs note

A hard check failing means NO-GO. A note is reported but does not block — used where the criterion is a preference rather than a limit.

Pass / fail counts — "25 passed, 1 failure"

25 individual checks returned yes; 1 returned no. Not a percentage: the one failure is enough for NO-GO regardless of the 25.

2. Geometry and CAD

STEP

The exact CAD file. It stores real surfaces — this face is a cylinder of radius 21, axis along Z. You can ask a STEP file questions: "is there a Ø42 hole here?" Used for every verification in this project.

STL

A mesh — the shape approximated by triangles. This is what the printer reads. It cannot tell you "this is a cylinder", only "here are 20 000 triangles". Used for collision checks and printing.

B-rep (boundary representation)

The way a STEP file describes a solid: as the set of surfaces that bound it. The reason a STEP can be interrogated and an STL can only be measured.

Boolean

A CAD operation that combines shapes: union (add), cut (subtract), intersect.

Silent no-op

A boolean that succeeds while doing nothing. Subtract a Ø42 pocket from a region that is already empty air and the operation succeeds and changes nothing — no error, no warning. This produced three missing bearing pockets in ARM-450 while a parameter check reported 55/55 passing. It is the single most important failure mode in the project.

Solid / body

One connected lump of material. A part should be exactly one. A .union() of two shapes that do not touch also "succeeds" — it just returns two separate solids. That is how the docking probe came to have three latch lugs floating in mid-air.

Watertight

A mesh with no holes in its surface — a sealed shell. A non-watertight mesh has no well-defined inside, so its volume is meaningless and a slicer may refuse it.

Fillet

A rounded internal or external corner. Removes the stress concentration a sharp corner creates.

Chamfer

A flat 45° cut across an edge. Not decorative on a bearing pocket: it is the lead-in that lets a Ø42 bearing start square in a Ø42 hole. ARM-450 had these specified everywhere and cut nowhere until 2026-08-25.

3. Fits and fasteners

Interference / press fit

The hole is smaller than the part going in, so it grips. A bearing in its pocket.

Clearance / slip fit

The hole is larger, so the part slides in. The horn adapter on the shaft — Ø30.20 on a Ø30 shaft — where a roll pin, not the fit, carries the torque.

Fit coupon

A small test print that measures what your machine does to a critical dimension. ARM-450's carries three bearing pockets at Ø41.95 / 42.00 / 42.05, three insert-hole sizes and a seam sample. It prints in 0.9 h and gates roughly 19 h of parts — press a bearing into each and whichever gives a firm thumb press is the right diameter for your printer.

Preload

Deliberate clamping force applied before any working load. The M5 through-bolt squeezes the two bearings of a pair against each other, removing internal play. Without it the joint has slack even with perfect bearings.

Heat-set insert

A brass threaded sleeve pressed into plastic with a soldering iron. Gives a real metal thread in a printed part — a screw threaded straight into PLA strips after a few cycles.

Grip length

How much material a screw passes through before it reaches the thread. Getting it wrong is invisible: an M3 × 12 in ARM-450's seam looked fully seated and engaged nothing, because it stopped 2.5 mm short of the insert.

Roll pin / spring pin

A slotted, slightly oversized pin that compresses on the way in and springs outward. Zero clearance, therefore zero backlash — which matters on an arm already backlash-limited.

4. Kinematics

DOF (degrees of freedom)

Independent ways to move. ARM-450 has 6 — the minimum to place a tool at any position and any orientation in space.

Forward kinematics (FK)

Joint angles → where the tool is. Always has exactly one answer.

Inverse kinematics (IK)

Where you want the tool → what joint angles get it there. The hard direction: there may be several answers, or none.

DH table (Denavit-Hartenberg)

A standard four-numbers-per-joint recipe for describing a robot's geometry. Compact and unambiguous, so two engineers with the same table build the same maths. ARM-450 uses the modified (Craig) convention.

Jacobian

The matrix relating joint speeds to tool speed. "If I turn J2 at 1 rad/s, how fast and in what direction does the tool move?" Used by IK to decide which way to step.

Velocity propagation

A way of building the Jacobian by carrying velocity from one link to the next, rather than by nudging each joint and measuring — which stays exact where numerical differencing degrades.

Singularity

A pose where the arm loses a direction of movement. In ARM-450 the J4 and J6 roll axes line up when J5 → 0, and only their sum matters — the solver can then wander wildly. Handled by locking J6 and damping the redundant pair.

TCP (tool centre point)

The point the robot is commanded to. ARM-450's is 450 mm from the base, at the J6 face.

Flange

The mounting face where a tool bolts on. Not the same as the working point: a gripper's fingers reach 42 mm past the flange — a 42 mm error if you confuse them, and this project made exactly that mistake.

Workspace

Everywhere the tool can reach. Quoted as maximum reach (450 mm bare) and swept volume (188 litres).

Waypoint

One point on a path. ARM-450's sine trace is 240 of them.

5. Structural analysis

FEA (finite element analysis)

Chop the part into small blocks, solve for the stress in each. Handles shapes too complicated for a formula.

Stress (σ)

Force per unit area, in MPa. How hard the material is being pulled or pushed internally.

Yield

The stress at which the material stops springing back and stays bent. PLA $\approx$ 55 MPa.

Safety factor (SF)

Yield $\div$ working stress. How many times harder you could push before it deforms. ARM-450's worst is 23 — meaning strength was never the binding constraint.

Compliance

The opposite of stiffness — how far the tool moves under load. ARM-450: 0.197 mm at full extension with 300 g.

Creep

Slow permanent sag under a load held for a long time. Plastics do this at room temperature; metals mostly do not. ARM-450's is 0.097 %/year — hence park the arm folded when idle, not extended.

Fatigue

Failure from many small load cycles, each far below what would break it once.

S-N curve

The stress-vs-number-of-cycles relationship. Metals often have an endurance limit — a stress below which they last forever. Polymers do not, so ARM-450 uses a life target (10^8 cycles) instead.

Backlash

Lost motion — the slack you feel when you reverse a gear before anything moves. In ARM-450 it is $\approx$ 2.24 mm at the tool and dominates everything: it beats structural deflection by about 20 $\times$.

If you remember one term, make it this one. The whole precision story of this project is that the arm is servo-limited, not structure-limited — the plastic is not the problem.

rms (root mean square)

A typical error across many points, giving more weight to large ones. Reported alongside max because a path can have a good average and one bad excursion.

6. ROS and simulation

ROS 2

The robot software framework. Programs are nodes; they communicate by publishing and subscribing to named topics.

Node

One running program. ARM-450 has `sine_node` (plays the trajectory) and `robot_state_publisher` (works out where every link is).

Topic

A named channel. `/joint_states` carries the six joint angles.

URDF

The robot's description file — links, joints, axes, limits, meshes, masses. Everything that needs to know the robot's shape reads it.

TF

ROS's system for tracking where every frame is relative to every other, continuously.

RViz

The 3-D viewer. It only draws what the URDF says — which is why an arm can be verified with a gripper fitted and still display without one, as ARM-450's did.

Latched (TRANSIENT_LOCAL)

A message kept for late subscribers. A controller that starts after the trajectory was published still receives it.

Collision-free

No two parts of the arm occupy the same space at any point on the path — checked on the real triangle meshes, not simplified boxes. ARM-450: 0 of 240 waypoints.

Exact mesh collision

Checking with the actual part geometry. The cheap alternative is bounding boxes, which would have missed the 1 mm clearance at the base pedestal.

7. Printing

Nozzle

The hole the plastic comes out of. 0.4 or 0.6 mm.

Layer height

How thick each printed layer is. 0.20 mm for ARM-450, and not a preference: the seam tongue is 1.6 mm and must be a whole number of layers, or the slicer rounds it up and the tongue bottoms out before the two halves can close.

Perimeter / wall line

The outlines the printer traces round each layer. Four perimeters $\times$ 0.62 mm = 2.48 mm, which consumes ARM-450's 2.4 mm wall entirely — that is what makes it solid.

Infill

The internal lattice filling hollow space. 40 % typical; 100 % where teeth are 2 mm, or they print as hollow shells and shear off.

Shrinkage

Plastic contracts as it cools, so a printed hole comes out smaller than drawn. Plain PLA 0.40 %, carbon-filled 0.15 % — 0.17 mm on a Ø42 bore, which is why the coupon exists.

XY compensation

A slicer setting that grows or shrinks every hole to correct for shrinkage. Set from the coupon result, not guessed.

Support

Scaffolding printed under overhangs. ARM-450 uses none on load paths — support leaves a rough weak surface, and you do not want that on a bearing seat.

Brim

A flat skirt around the first layer for grip. Used on tall narrow parts.

8. This project's own vocabulary

Interrogate the geometry

Ask the exported STEP or STL what it contains, rather than trusting the numbers used to draw it. The central rule of the whole verification method.

Parameter check vs geometry check

A parameter check reads BEARING_POCKET_D == 42.0 — your intention. A geometry check opens the file and looks for a Ø42 cylindrical face — the result. They disagreed here, and the parameter check was the one that passed.

Capture tolerance

How far off you can be and still succeed. ARM-450's docking cone accepts ± 9.75 mm against ≈ 2.2 mm of arm error — docking made feasible by geometry rather than by

accuracy.

Bayonet

A push-and-twist coupling — like a camera lens. Used twice here: the tool quick-change, and the docking latch that J6 turns.

Payload vs arm mass

Payload is what the arm carries; arm mass is the arm itself. A fitted tool is payload — which is why a 75 g gripper does not count against the 1450 g budget, but 244 g of bolts does.

Falsification condition

What result would prove the prediction wrong, written down before the test. "Backlash 1.9–2.2 mm; falsified if under 0.5 mm." Without it a prediction can be quietly retro-fitted to whatever happens.